

Code Fix Remote



DESCRIPTION

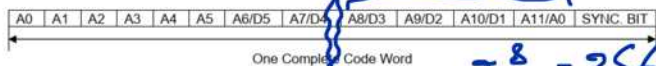
PT2262 is a remote control encoder paired with PT2272 utilizing CMOS Technology. It encodes data and address pins into a serial coded waveform suitable for RF or IR modulation. PT2262 has a maximum of 12 bits of tri-state address pins providing up to 531,441 (or 312) address codes; thereby, drastically reducing any code collision and unauthorized code scanning possibilities.

FEATURES

- CMOS Technology
- Low Power Consumption
- Very High Noise Immunity
- Up to 12 Tri-State Code Address Pins
- Up to 6 Data Pins
- Wide Range of Operating Voltage: $V_{cc} = 4 \sim 15V$
- Single Resistor Oscillator
- Latch or Momentary Output Type
- Available in DIP and SOP

$$2^{12} = 4096$$

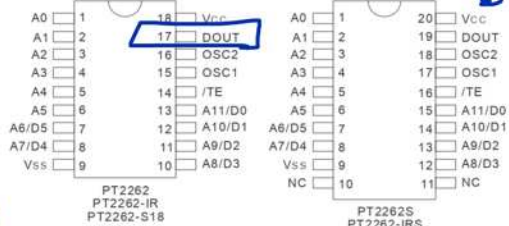
$$2^8 = 256$$



PT2262 / PT2272 has a maximum of twelve (12) Address Bits including the six (6) Address/Data bits. The following diagram shows the code bits with their corresponding pins.

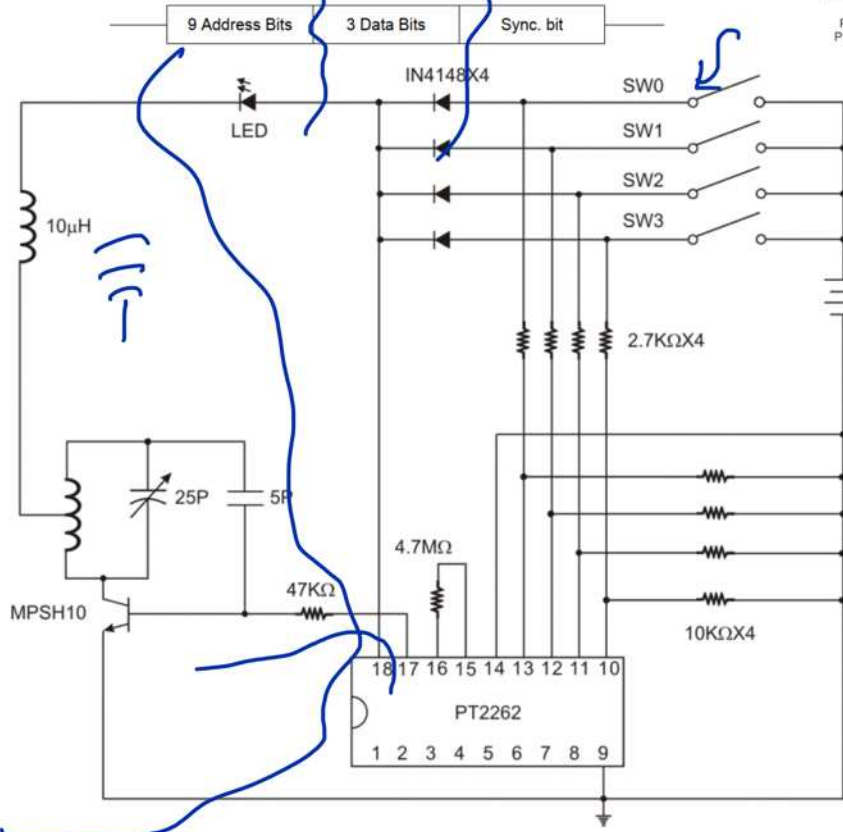
0 Data:	A0	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	Sync Bit
1 Data:	A0	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	D0	Sync Bit
2 Data:	A0	A1	A2	A3	A4	A5	A6	A7	A8	A9	D1	D0	Sync Bit
3 Data:	A0	A1	A2	A3	A4	A5	A6	A7	A8	D2	D1	D0	Sync Bit
4 Data:	A0	A1	A2	A3	A4	A5	A6	A7	D3	D2	D1	D0	Sync Bit
5 Data:	A0	A1	A2	A3	A4	A5	A6	D4	D3	D2	D1	D0	Sync Bit
6 Data:	A0	A1	A2	A3	A4	A5	D5	D4	D3	D2	D1	D0	Sync Bit

For example: In the 3 Data Type where the address has nine (9) bits, the transmitting format is:



2-BUTTON Design

The universal remote can even operate up to two different brands of garage door openers



Code Learn Remote



EV1527 Description: (Compatible:EV1527,RT1527,FP1527)

EV1527 is OTP Encoder utilizing CMOS technology process. EV1527 hai a maximum of 20 bits providing up to 1 million codes.It can reduce any code collision and unauthorized code scanning possibilities.

EV1527 Features: (Compatible:EV1527,RT1527,FP1527)

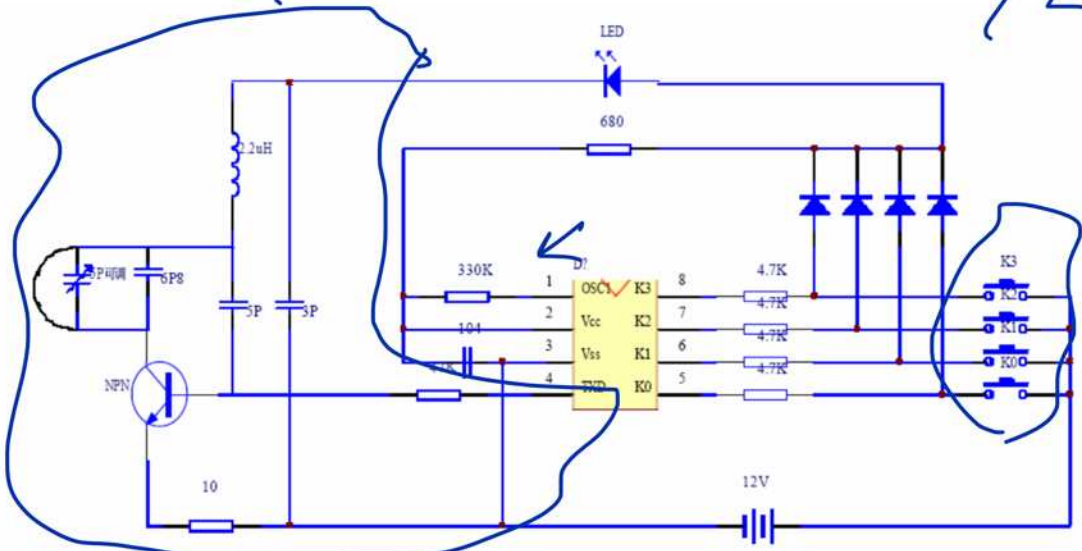
- CMOS Technology
- Low stand by Current:1.0uA
- Wide range of Operating Voltage:Vcc=3.0~12V
- Up to 4 data pins
- Total 1048576 address codes
- Single Resistor Oscillator
- Available in DIP and SOP Package



Output data frame:

Preambl e	C0 ~ C19 (1 million codes) $2^{20} = 1,048,576$	D0	D1	D2	D3
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EV1527 Schematic diagram:(Compatible:EV1527,RT1527,FP1527)



$\frac{1}{256} \approx \frac{1}{11}$
433 MHz
315 MHz

EV1527 K0~K3 combination Table: (Compatible:EV1527,RT1527,FP1527)

K3	K2	K1	K0	D3	D2	D1	D0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	0
0	0	1	1	0	0	1	1
0	1	0	0	0	1	0	0
0	1	0	1	0	1	0	1
0	1	1	0	0	1	1	0
0	1	1	1	0	1	1	1
1	0	0	0	1	0	0	0
1	0	0	1	1	0	0	1
1	0	1	0	1	0	1	0
1	0	1	1	1	0	1	1
1	1	0	0	1	1	0	0
1	1	0	1	1	1	0	1
1	1	1	0	1	1	1	0
1	1	1	1	1	1	1	1

Remote Control Duplicator

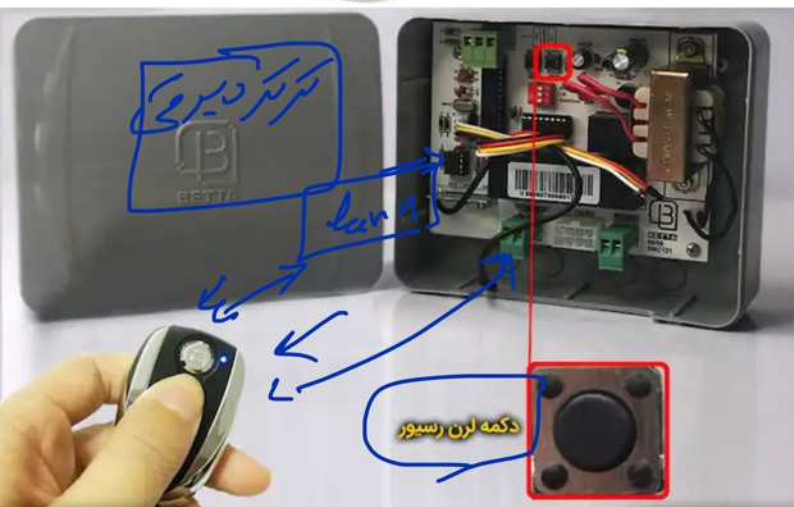
بلوتوث



20-Address
4-Data

D1	1	16	D0
D2	2	15	PGM
D3	3	14	SIO
VSS	4	13	VPP
NC	5	12	VDD
NC	6	11	DOUT
NC	7	10	OSC1
NC	8	9	OSC2

HT6P20D
16 DIP/NSOP



Each Button Supports 280-900mhz



280mhz to 900mhz

280mhz to 900mhz

280mhz to 900mhz

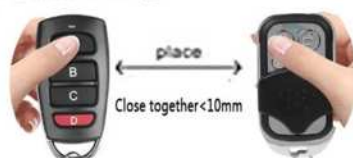
280mhz to 900mhz

1.operation

The Original Remote Control must be able to work and the same as our remote control frequency. (e.g. 433MHz => 433MHz)



3.Code cloning



Press the 2 buttons at the same time when the led indicators flash all the time then release them. (Close together <10mm)

Tips: press the cloned-button on the clone remote. If the led indicator turns on all the time that means that the code is cloned successfully. Clone the other codes from the original remote by following the step "3"

Learning frequency range:

287MHz, 300MHz, 306MHz, 310MHz, 315MHz, 315.5MHz, 330MHz, 360MHz, 390MHz, 410MHz, 430.5MHz, 433.42MHz, 433.92MHz, 868.35MHz, 286-361MHz, 384-400MHz.

2.Code clear



Press the A & B buttons at the same time until the LED flashes, then release the A button and press A button 3 times. (Meanwhile, button B must be pressed all the time.)

Press any button, the light is off then "Code clear success".

4.Code recovery



Press and hold the two buttons (C & D) under the copy remote control at the same time until the indicator light flashes quickly, you will restore the remote control code that was cleared before.



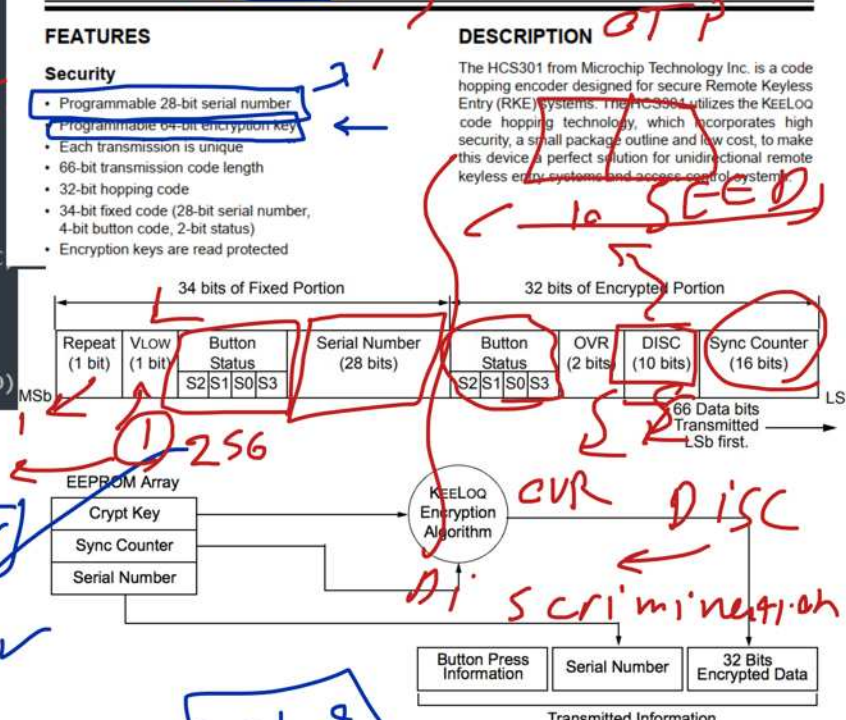
1 million = 4x

کد کپی ۱
و کد رسیور



Address: 0x59647, Control: 0x1
 Address: 0x59647, Control: 0x1
 Address: 0x59647, Control: 0x1
 Address: 0x59647, Control: 0x1
 Address: 0x59647, Control: 0x1
 Address: 0x59647, Control: 0x1
 Address: 0x59647, Control: 0x1
 Address: 0x59647, Control: 0x1
 Address: 0x59647, Control: 0x1
 Address: 0x59647, Control: 0x1
 Address: 0x59647, Control: 0x1

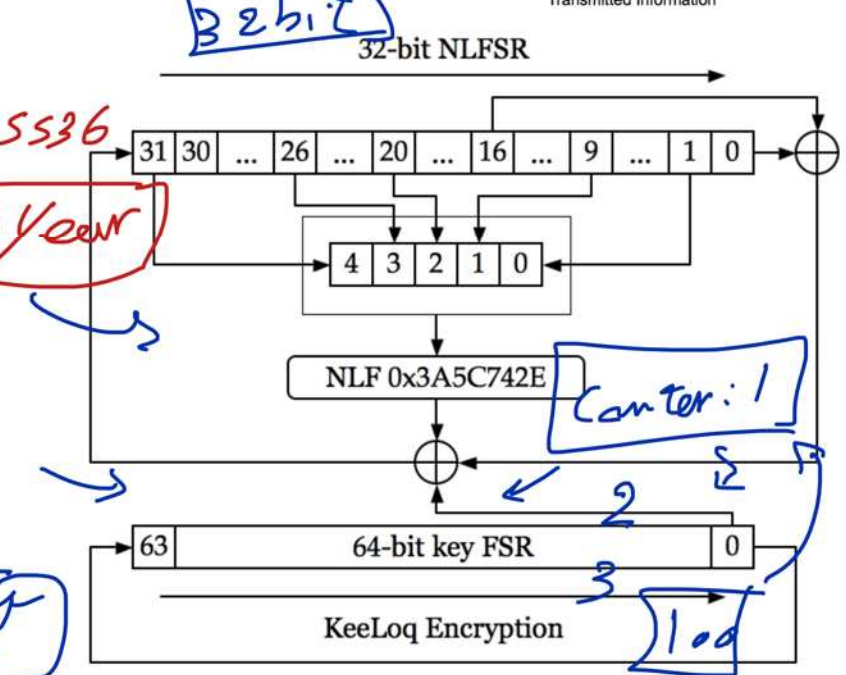
0xA0C18A8A3656690C
 0xA0C18A8A46DDACD
 0xA0C18A8A498D209E
 0xA0C18A8A498D209E
 0xA0C18A8ADC76F1E3
 0xA0C18A8A939C0EFA
 Repeat: 0
 Vlow : 0
 Button: 0b0010 (S1)
 Serial: 20C18A8A
 OVR : 0b11
 DISC : 0x28A
 0xA0C18A8A090EACF6
 Fix: A0C18A8A
 Enc: 090EACF6
 Dec: 2E8A087C (Counter=087C)
 0xA0C18A8A498D209E
 Fix: A0C18A8A
 Enc: 498D209E
 Dec: 2E8A087D (Counter=087D)

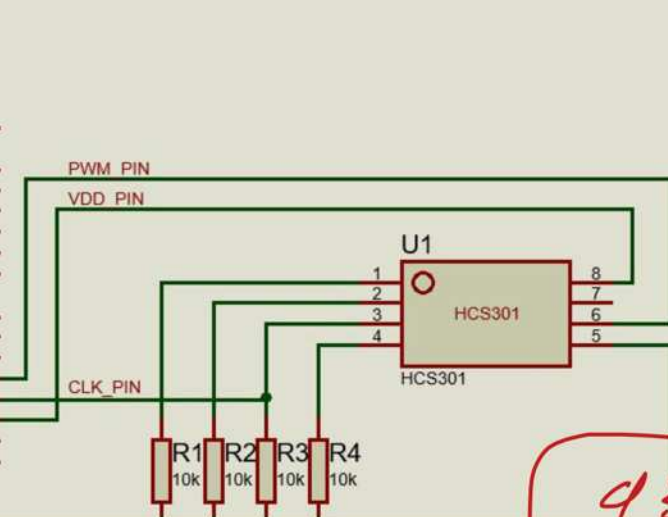
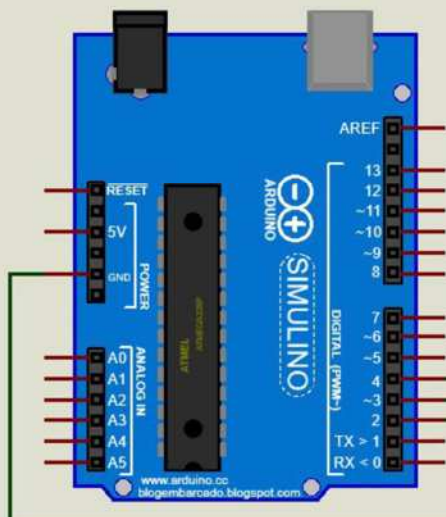
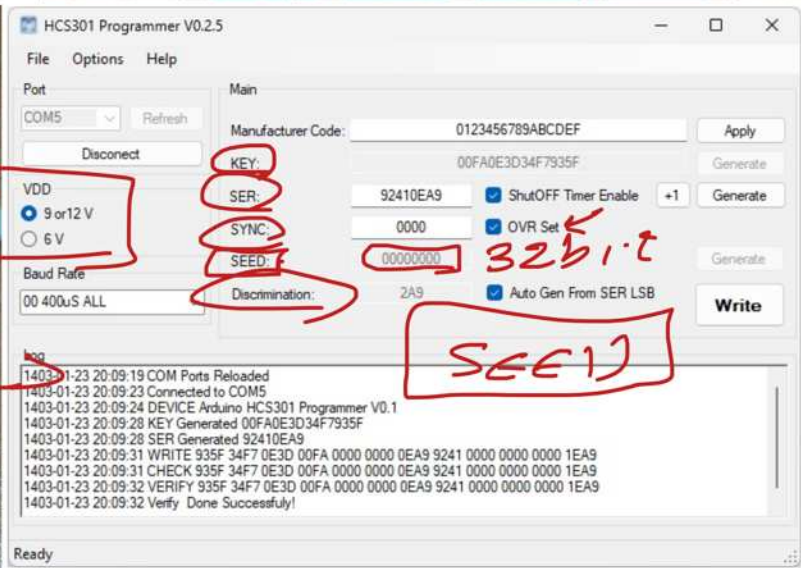
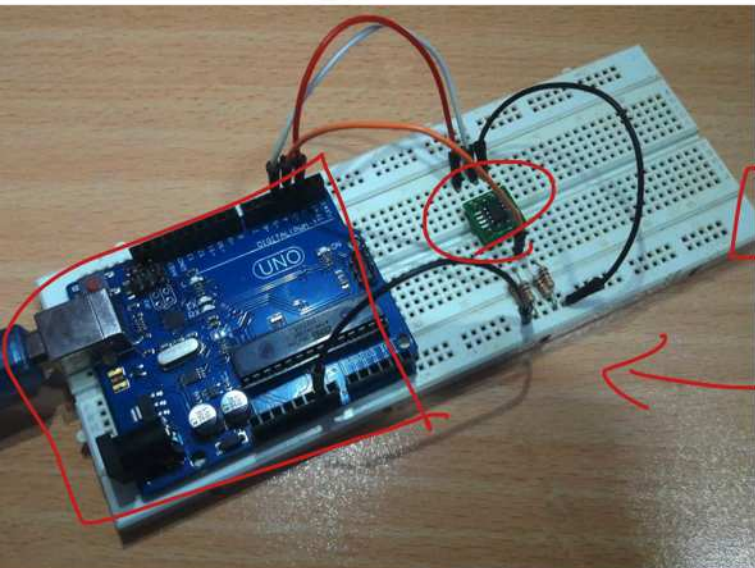
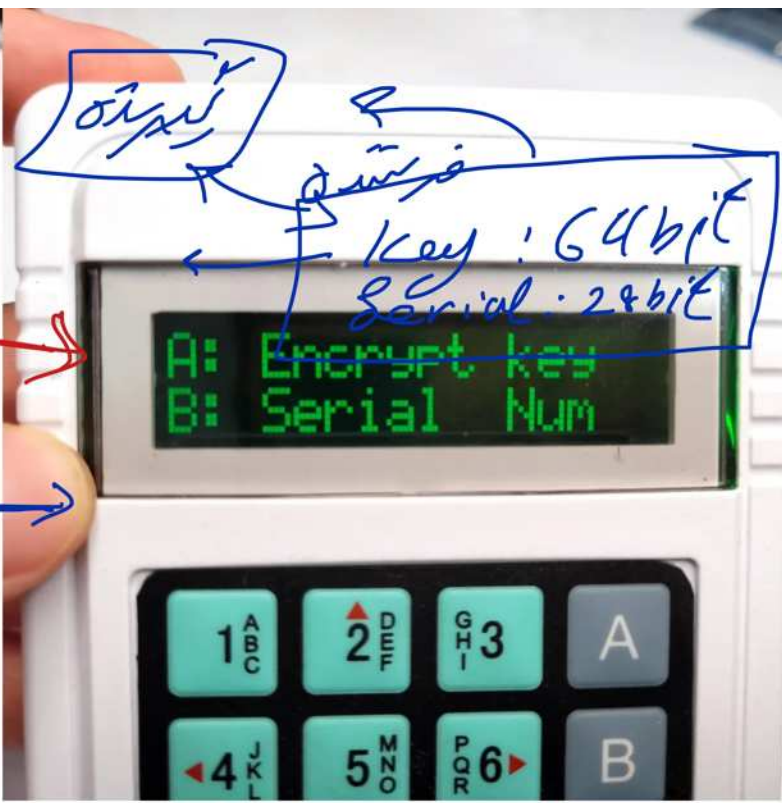


RF 433
 RF 315
 code fix

1 million
 256
 learn
 counter
 computer

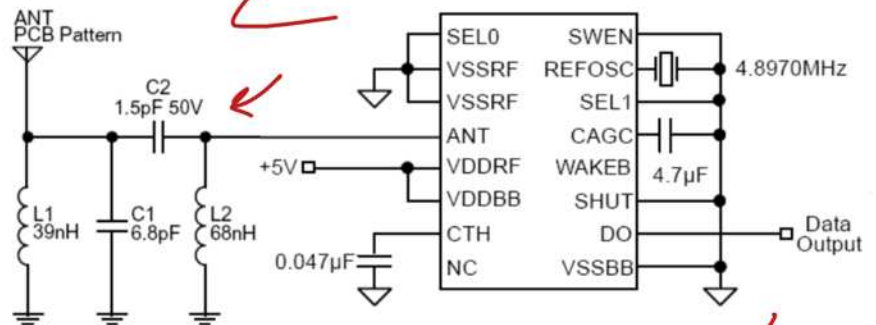
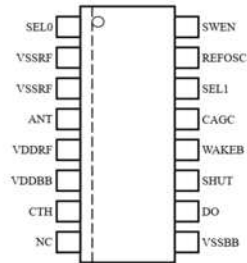
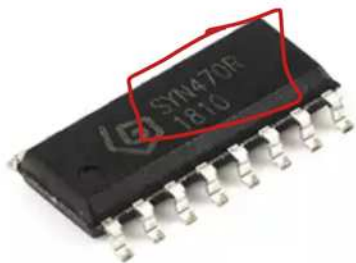
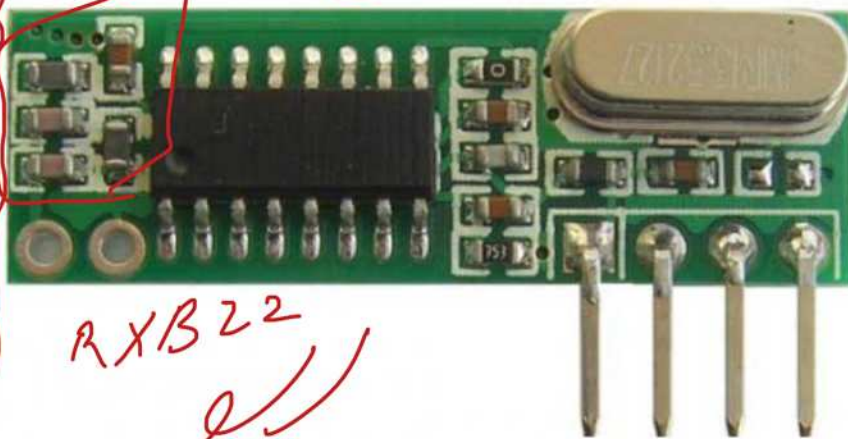
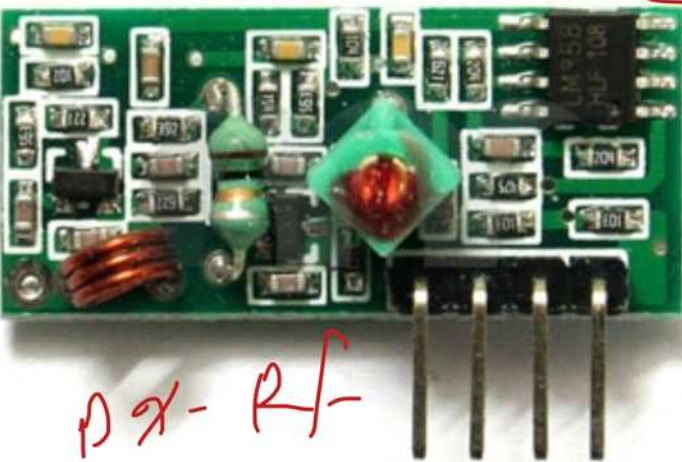
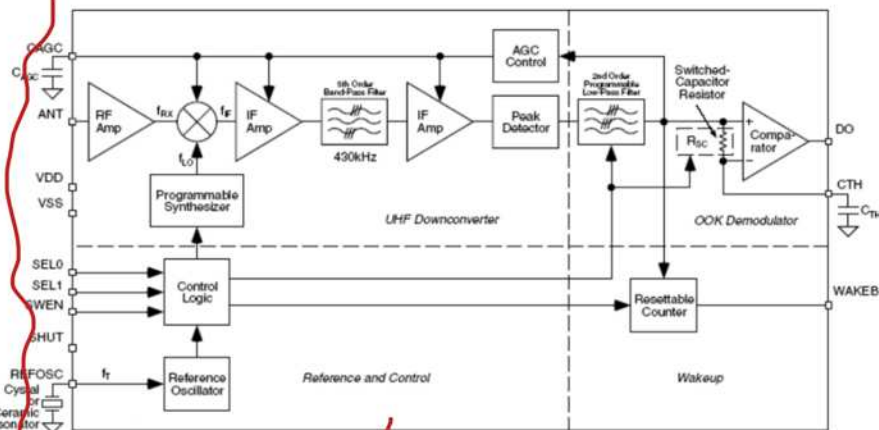
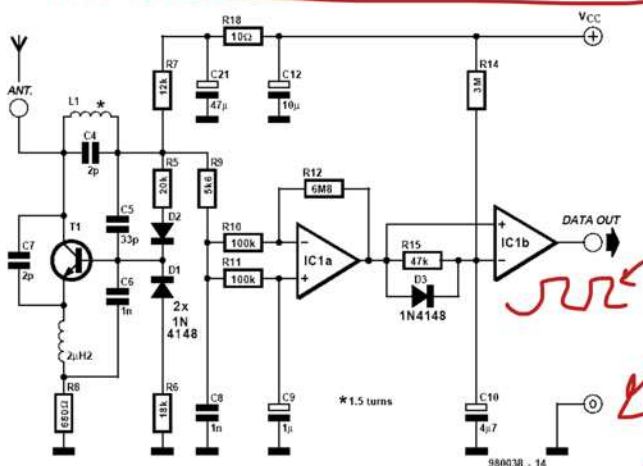
learn 1 million
 2¹⁶ = 65536
 18 years
 counter → 1
 → 2
 → 3
 Serial & counter





433MHz
315MHz

Super Regenerative vs SuperHeterodyne



SEL0	SEL1	Demodulator Bandwidth	
		Sweep Mode	Fixed Mode
1	1	5000 Hz	10000Hz
0	1	2500 Hz	5000Hz
1	0	1250 Hz	2500 Hz
0	0	625 Hz	1250 Hz

Transmit Frequency (f_{TX})	Reference Oscillator Frequency (f_T)
315MHz	4.8970 MHz
390 MHz	6.0630 MHz
418 MHz	6.4983 MHz
433.92 MHz	6.7458 MHz

Poor anti-interference ability

The frequency is greatly affected by temperature drift

Low receiving sensitivity

Frequency pollution, can not pass the certification

Low price, poor stability

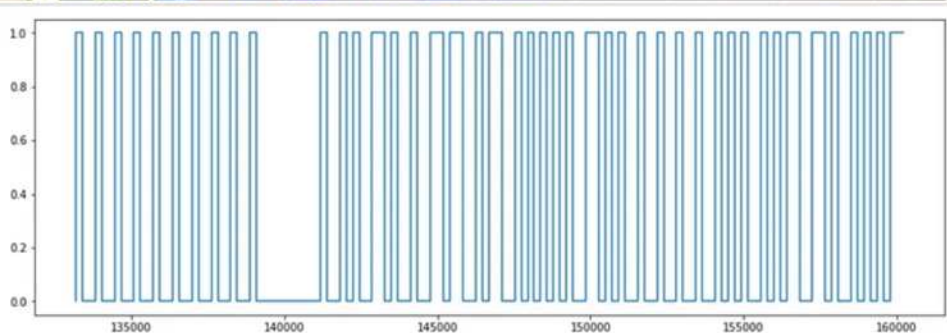
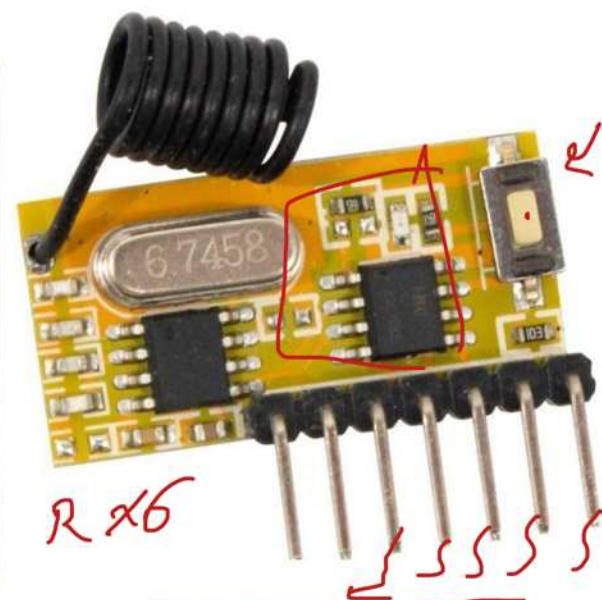
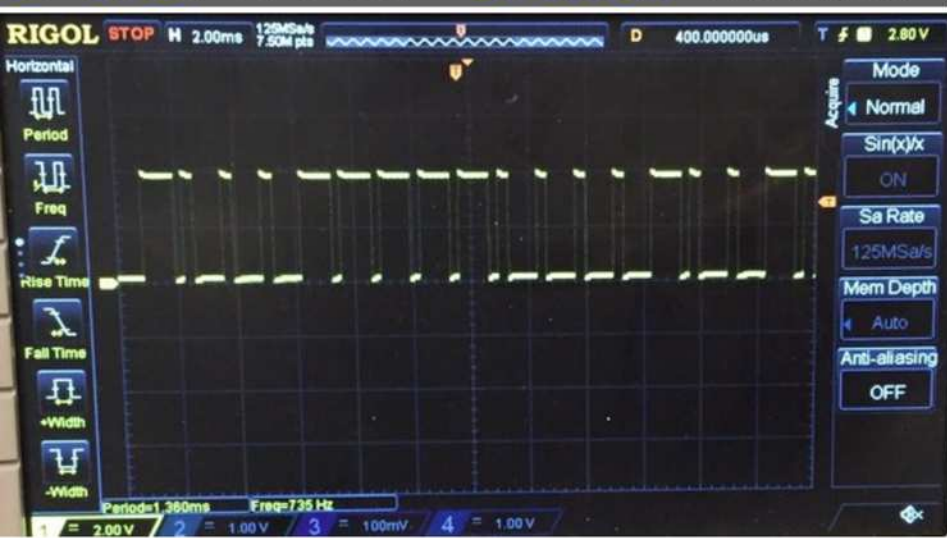
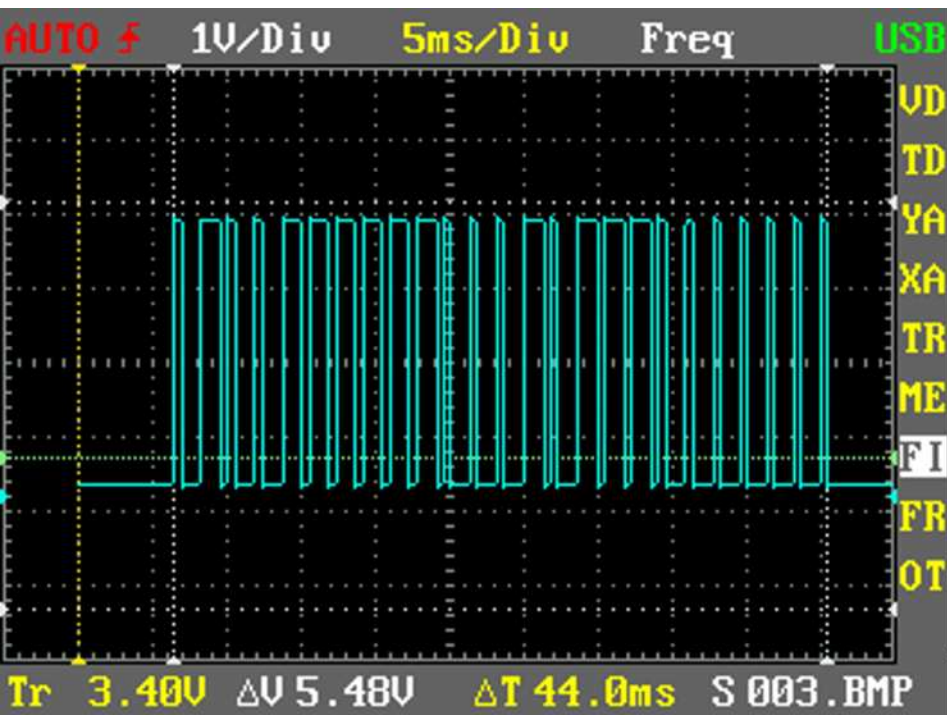
Strong anti-interference ability

Strong temperature adaptability

High receiving sensitivity

Able to pass FCC, CE certification

Good performance, stable and reliable work





Type	Typical IC	Encoding Method	Security	Price
Fixed Code	PT2262	Manual Encoding, using dip switch or soldering to set address code	Low	Low
Learning Code	EV1527	Automatic Encoding, copying fixed code or learning code remote	Medium	Medium
Rolling Code	HCS301	Automatic Encoding, using algorithm to generate different code every time	High	High

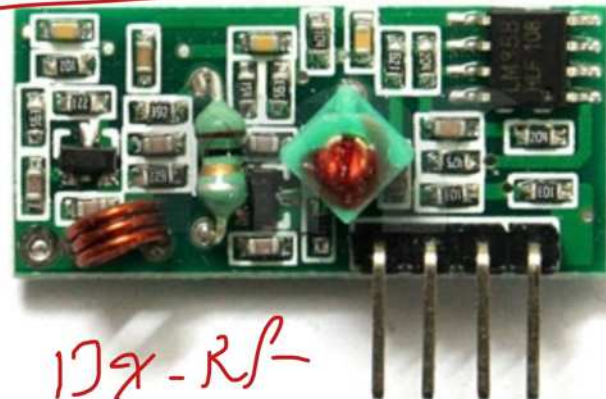
Chip Model	Fixed Code	Rolling Code	Learning Code
[2240] PT2240 [2242] PT2242 [2260] PT2260 [2262] PT2262 / M3E / CT5062 [6010] HT6010 / HT6012 / HT6014 / SH312E [H600] HT600 / HT680 / HT6187 / HT6270 / TT13 / PK10T [5026] AX5026 / CT5026 / SMC5026 [5326] AX5326 / AX5326S [8092] TT8092 [4134] MC4134 / MC1450 / SC4134 [918] SMC918	X		
HCS301 / HCS300 / HCS200		X	
EV1527 / FP527 / HT6P20B / HT6P20D			X



eV1527



RX B22



10x-RS-